AP Chemistry: Atomic Theory I & II **Bartonian ID _____**

Write neatly ◦ Boxes are reserved for *final answers only* ◦ *Bartonian ID* must be written on each page ◦ Do not staple.

1) What is an isotope?

1

Questions 2 – 5 refer to the $^{238}_{92}U$ isotope.

2) How many protons are contained within the nucleus?

2

3) How many neutrons are contained within the nucleus?

3

4) How many electrons are contained within a single neutral isotope of this element?

4

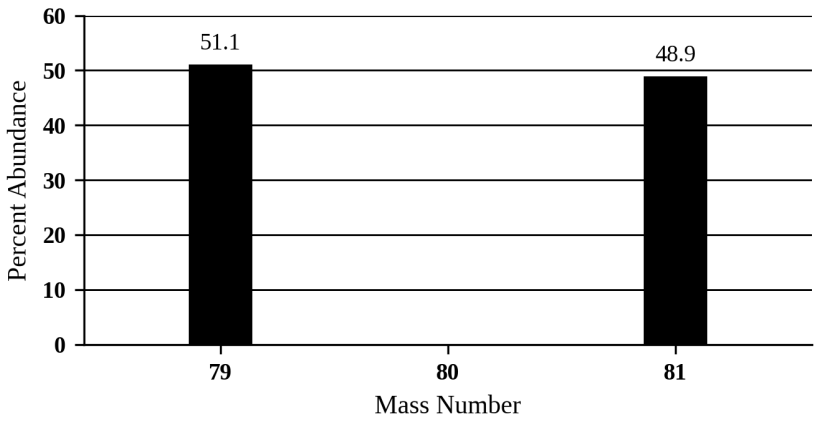
5) What is the mass of this isotope in amu?

5

6) Which isotope is more likely to bond with oxygen, ^{12}C or ^{14}C? Explain.

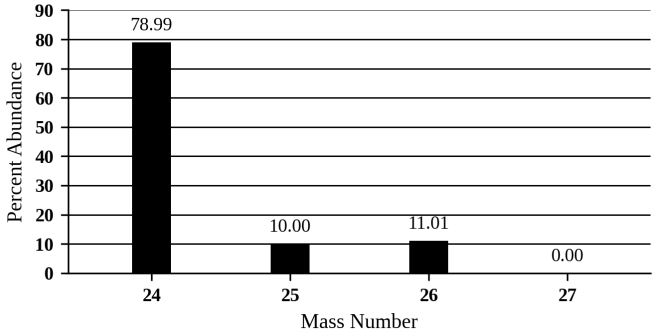
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7) A pure sample of bromine was vaporized and injected into a mass spectrometer and the data was plotted on the graph below. The mass value for Br-79 is 78.918 amu. Find the mass of Br-81.



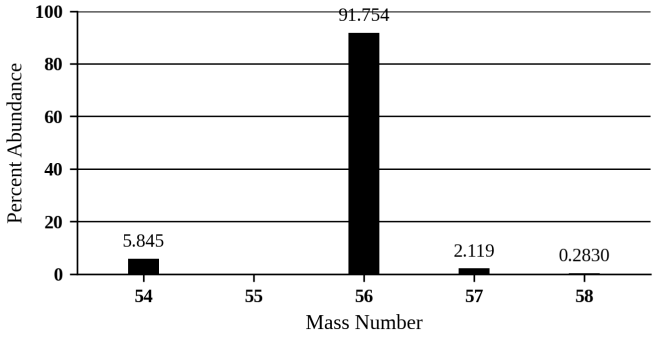
7

8) A pure sample of an element was vaporized and injected into a mass spectrometer and the data was plotted on the graph below. The mass values for the isotopes were found to be: A-24 (23.985 amu), A-25 (24.986 amu), and A-26 (25.983 amu). Find the average atomic mass and identify the element.



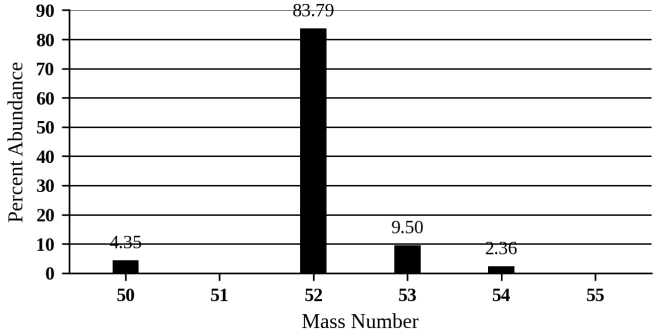
8

9) A pure sample of an element was vaporized and injected into a mass spectrometer and the data was plotted on the graph below. The mass values for the isotopes were found to be: A-56 (55.935 amu), A-54 (53.940 amu), A-57 (56.935 amu), and A-58 (57.933 amu). Find the average atomic mass and identify the element.



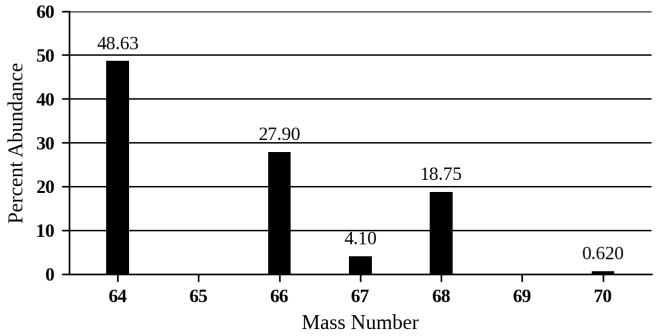
9

10) A pure sample of an element was vaporized and injected into a mass spectrometer and the data was plotted on the graph below. The mass values for the isotopes were found to be: A-50 (49.946 amu), A-52 (51.941 amu), A-53 (52.941 amu), and A-54 (53.939 amu). Find the average atomic mass and identify the element.



10

11) A pure sample of an element was vaporized and injected into a mass spectrometer and the data was plotted on the graph below. The mass values for the isotopes were found to be: A-64 (63.929 amu), A-66 (65.926 amu), A-68 (67.925 amu), A-70 (69.925 amu), and A-67 (66.927 amu). Find the average atomic mass and identify the element.



11

12) Calculate the mass percent of each element in $CaCO_3$.

12

13) Which colour of light has the highest frequency: red or green?

13

14) Which colour of light has the longest wavelength: green or violet?

14

15) Hydrogen emits light with a wavelength of 410 nm. Find the frequency of this light.

15

16) A radio station broadcasts at a frequency of 1310 kHz. What is the wavelength of this radio signal in nm?

16